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INSOMNIAC GAMES

Higher, Further, Faster

Evolving Traversal in Marvel's Spider-Man 2

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Previously On...



Traversal Pillars

- Spider-Man



Traversal Pillars

- Spider-Man
- Depth Through Momentum



Traversal Pillars

- Spider-Man
- Depth Through Momentum
- Breadth of Action



Traversal Pillars

- Spider-Man
- Depth Through Momentum
- Breadth of Action
- Broad Audience



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WEB WINGS





Design Goals

- Feel authentic but fantastical



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- Follow height-for-speed tradeoffs



Design Goals

- Feel authentic but fantastical
- Follow height-for-speed tradeoffs
- Interleave with existing traversal

Early Prototyping

- Start Fast
- Fail Fast
- Learn Fast

Early Prototyping

- Define attributes of authenticity
 - Gain speed as you dive
 - Lose speed as you glide
 - Gravity pulls you down
 - Steerable yaw and pitch
 - Limits on maximum speed
 - Limits on maximum pitch

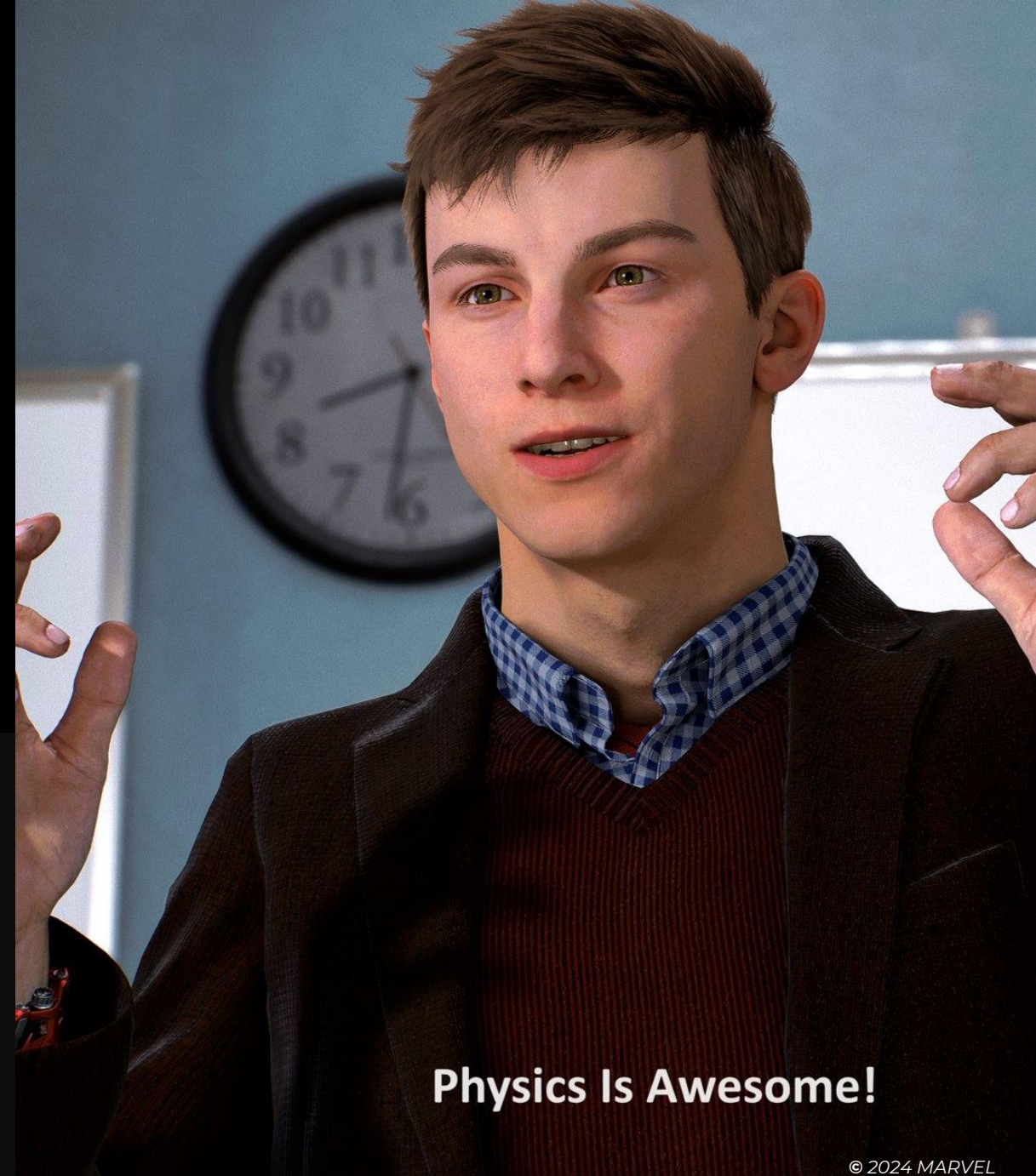
Failure



Early Prototyping - Results

- Failed to feel authentic
- Steering and speed were too linear
- Missing key elements of flight
- Not tunable

Back to Basics



Physics Is Awesome!

Wingsuit vs Glider

- Wingsuit visually
- Glider mechanically



Early Simplifications

- Body faces velocity

Early Simplifications

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- Input driven yaw/pitch

Early Simplifications

- Body faces velocity
- Input driven yaw/pitch
- Fixed body parameters

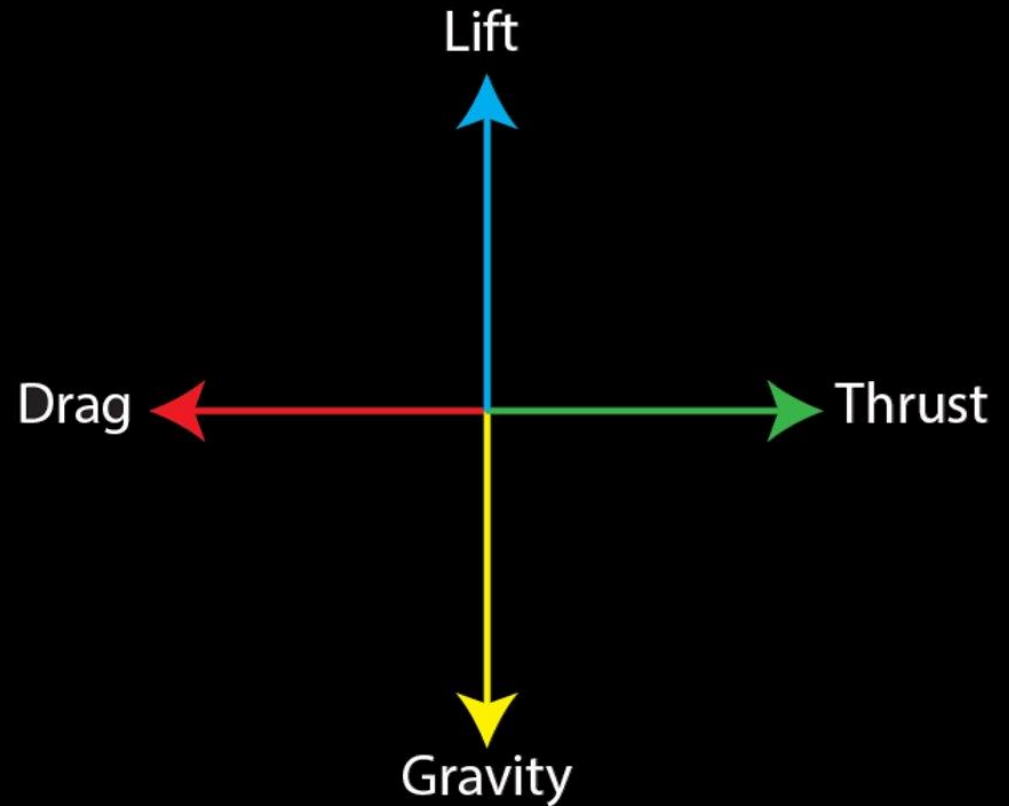
Glider Forces

Lift

Drag

Gravity

Thrust



Forces: Lift

F_L : Force of lift

C_L : Coefficient of lift

r : Air density

V : Velocity (speed)

A : Wing area

$$F_L = C_L * r * (V^2/2) * A$$

Ignore air density and wing area as non-variable, bake into coefficient

Forces : Drag

F_D : Force of drag

C_D : Coefficient of drag

r : Air density

V : Speed relative to motion

A : Facing surface area

$$F_D = C_D * r * (V^2/2) * A$$

Forces : Gravity

F_G : Force of gravity

F_G

Forces : Combined

V_F : Velocity Final

V_A : Velocity Average

$$A_L = F_L * D_L$$

$$A_D = F_D * |-V|$$

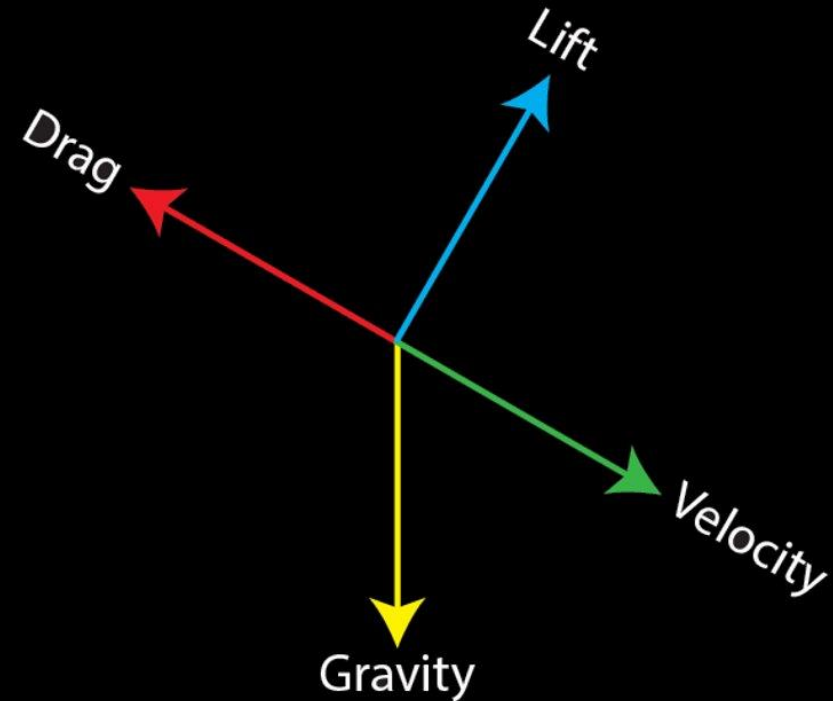
$$A_G = F_G * -U$$

$$V_F = V_I + (A_L * dt) + (A_D * dt) + (A_G * dt)$$

$$V_A = (V_F + V_I) / 2$$

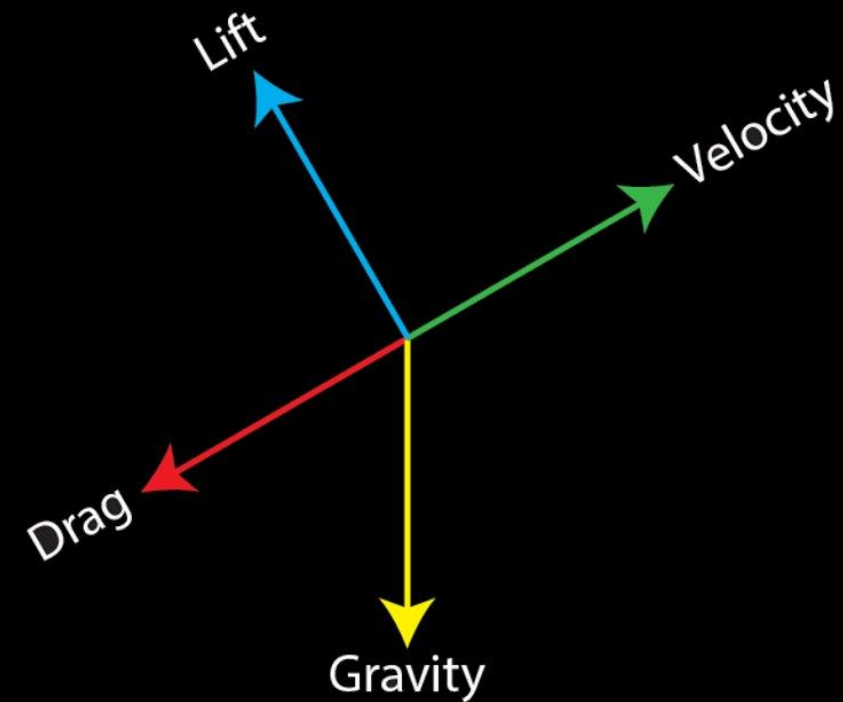
Forces : Combined

- Gravity increases overall speed
- Lift generates forward speed
- Lift counters fall speed



Forces : Combined

- Lift can act as a natural brake



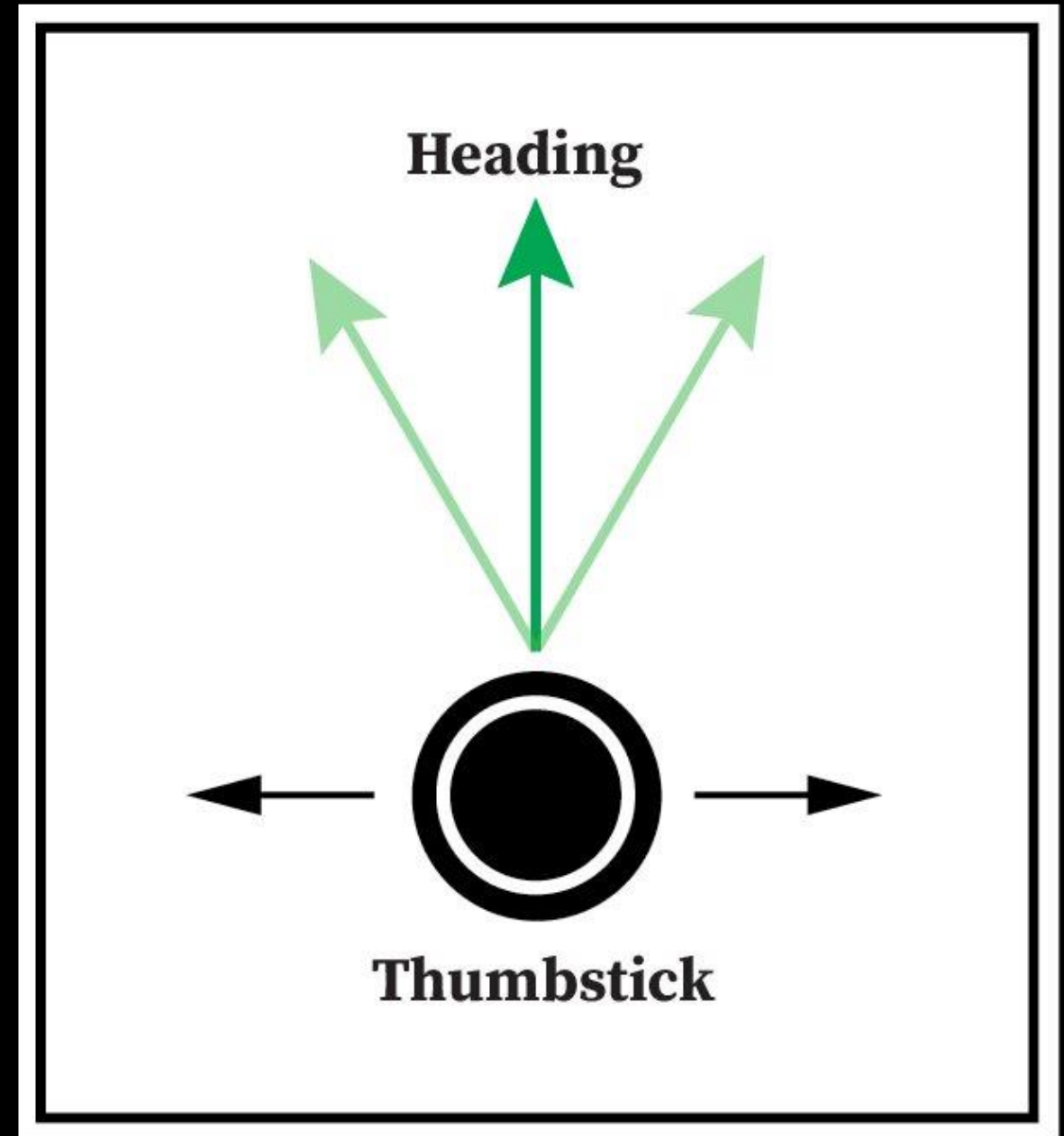
Player Input

- Left stick driven
- Full yaw control
- Variable pitch control



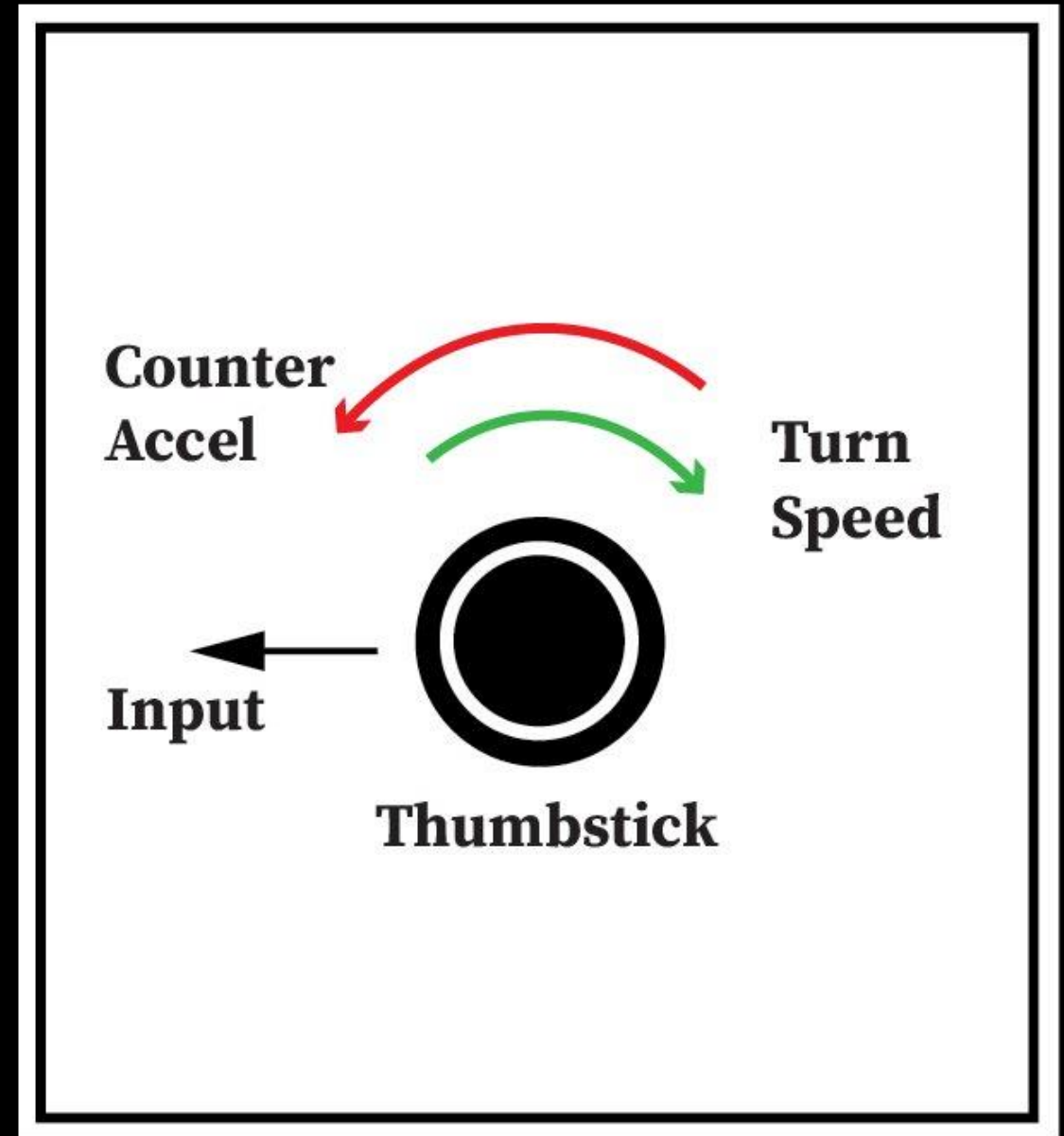
Yaw Control

- Use left/right stick strength



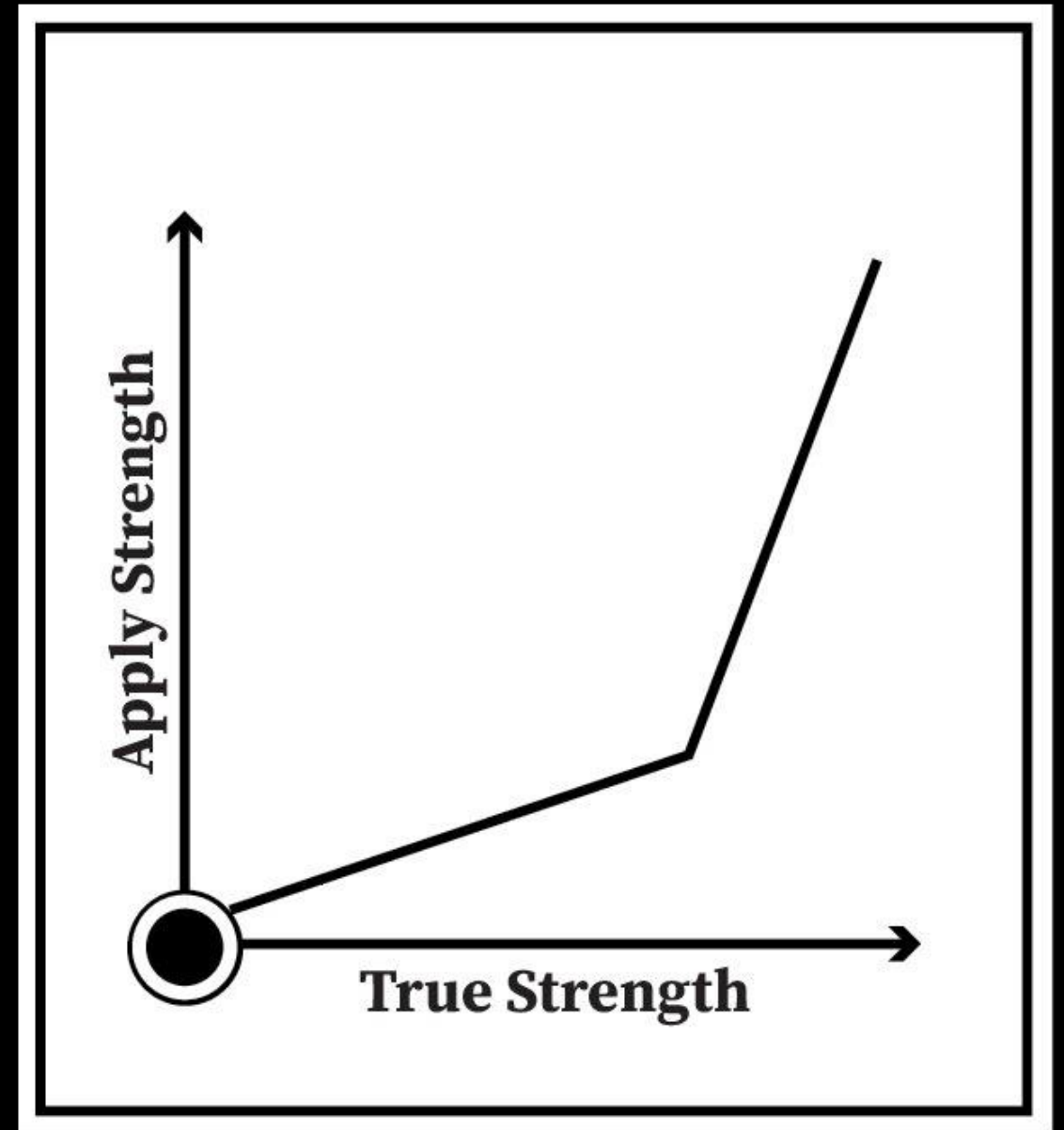
Yaw Control

- Use left/right stick strength
- Accel/Decel/Counter model



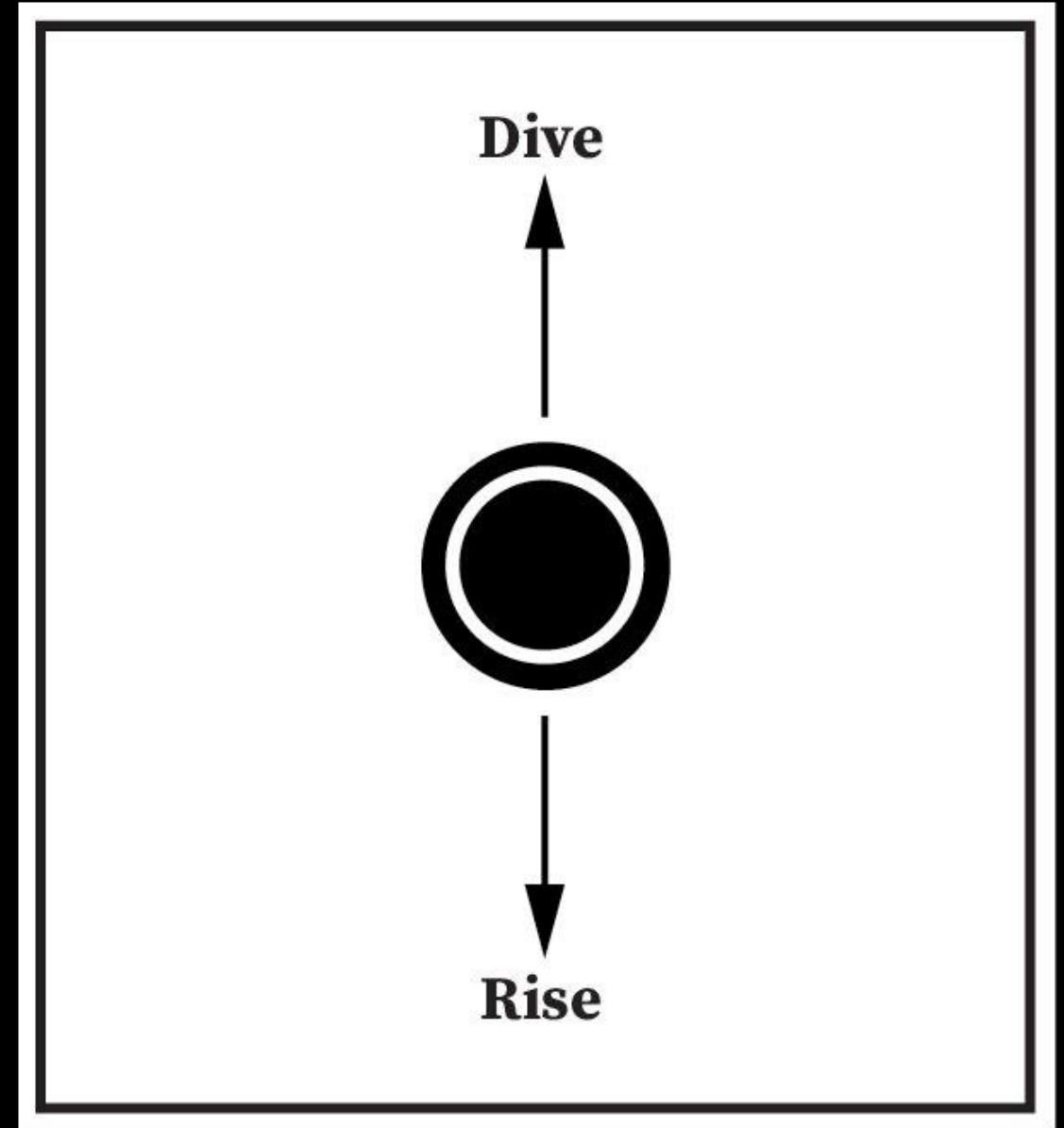
Yaw Control

- Use left/right stick strength
- Accel/Decel/Counter model
- Input ramp



Pitch Control

- Responsive
- Obey speed-vs-lift



Pitch Control

- Player can aggressively pitch down



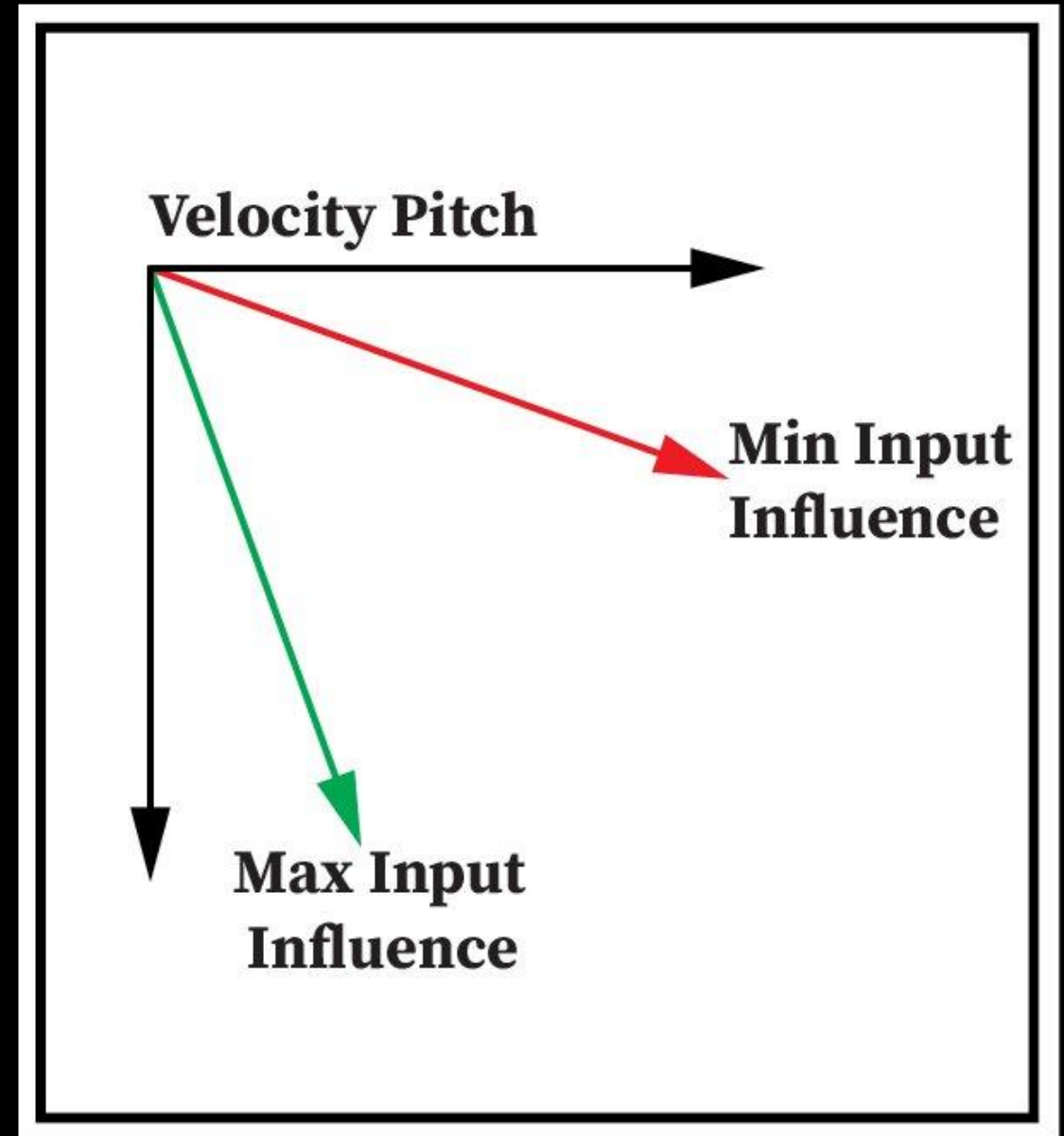
Pitch Control

- Player can aggressively pitch down
- Player can aggressively pull out of a dive



Pitch Control

- Player can aggressively pitch down
- Player can aggressively pull out of a dive – to a point



Pitch Control

- Player can aggressively pitch down
- Player can aggressively pull out of a dive – to a point
- Coefficient manipulation at and above level



PLAYABLE BY ALL

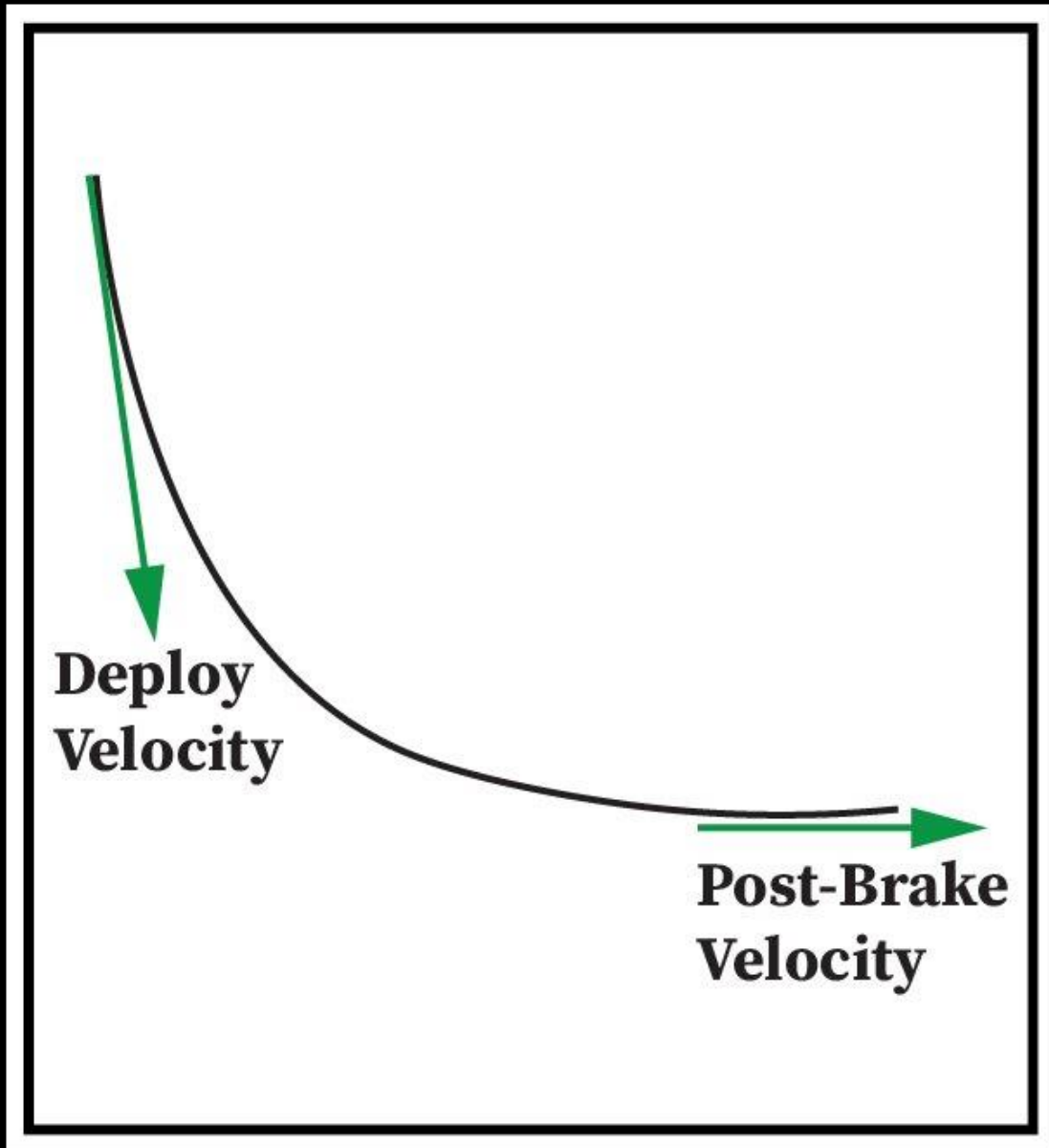


Deployment

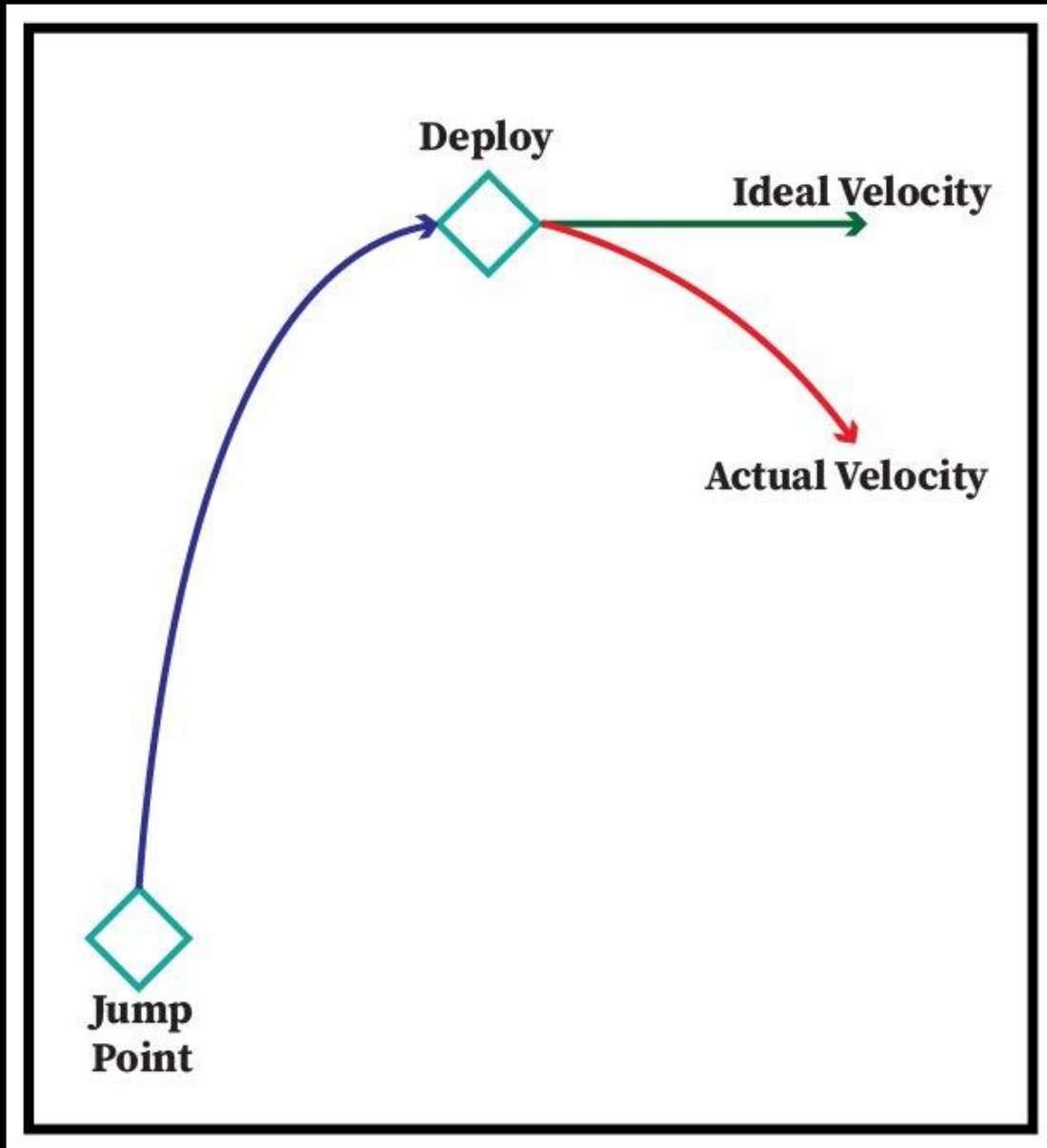
- R2 was our “go” button, contract with players
- Most of development was R2 + TRI
- UX consistently had difficulty
- Had to be willing to compromise design rules for playability
- Shipped with just tap of TRI to deploy

Playable By All : Deploy Vertical Brake

- Act as sort of drag chute
- Rapidly slow fall speed
- Convert to horizontal speed







Playable By All: Deployment Boost

- Lift requires speed
- Frequent sub-optimal deployment
- Give a boost to a minimum speed



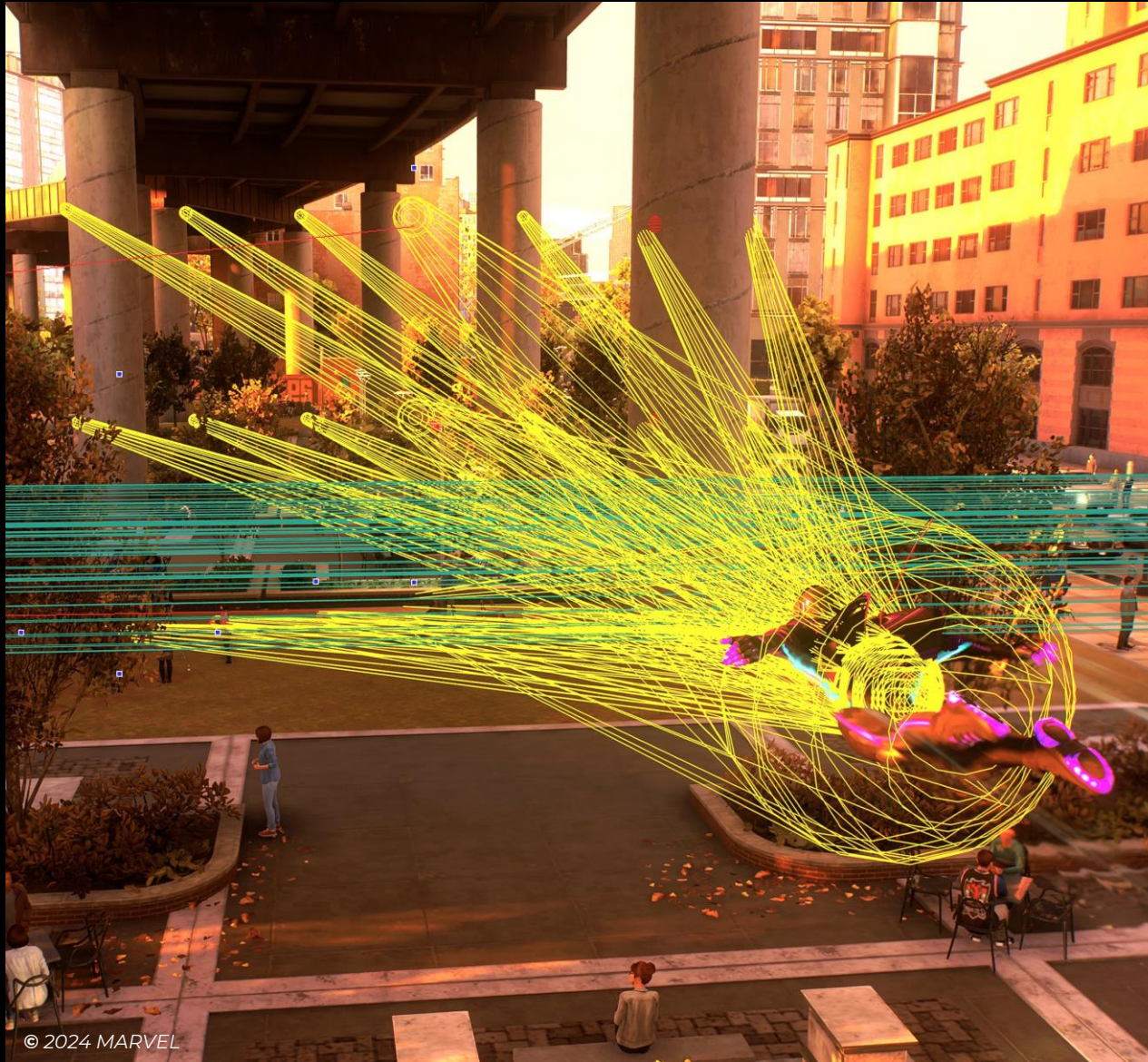
Playable By All: Flight Controls

- Flight controls vs standard
- Handle stick-back mode changes
- Built in stick-flip logic for transitions and targeting

Playable By All: Ground Assist



- Wanted feel of skimming ground
- Players were wiping out repeatedly
- Slight pitch steering bump near ground
- Created short duration anti-gravity bubble when near ground



Playable By All: Wall Assist

- Array of forward/side tests
- Nudge towards openings
- Dampen steering towards walls



Slipstreams

- Traverse city quickly
- Reward players for reading environment
- Break up experience cadence
- Maintain player control



SPEED BOOST

Gameplay Benefits

- Prior speed cap limited range of viable player speed
 - 22 – 30m/s
- Uncapping speed allows more natural motion
 - 45-60m/s base cap
 - 100m/s max speed
 - Fall speed can translate more directly to horizontal speed
- Increases skill ceiling as variety of moves can get meaningful rewards

Gameplay Challenges

- Too much speed left players out of control

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- Too much speed left players out of control
- Linearly ramped air drag proved insufficient
- Players build speed too fast

Air Drag

- Move to quadratic air drag model
 - Lower drag at lower speeds
 - Higher drag at higher speeds
- Applied on AND off the swing line

Momentum Meter

- Governs speed boost potential
 - Maximum speed gained in a single swing
 - Boost provided by a timely point launch

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- Governs speed boost potential
 - Maximum speed gained in a single swing
 - Boost provided by a timely point launch
- Never slows the player down
- Built through skillful traversal
- Decays over time



L1+R1

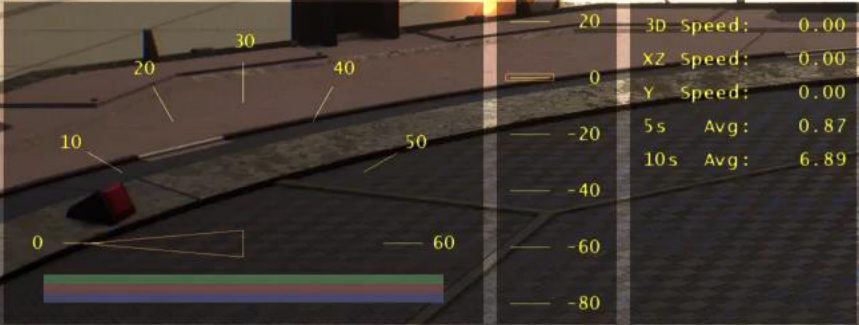


3D Speed:	0.00
XZ Speed:	0.00
Y Speed:	0.00
5s Avg:	1.76
10s Avg:	1.65



Aero Meter

- Many moves give player large speed boosts
- Speed is quickly eaten by air drag
- Building aero meter reduces drag
- Allows hero to hold speed for long durations



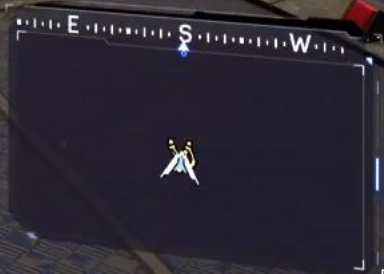
Hold **L2** + **X** Super Slingshot





20	3D Speed:	0.00
0	XZ Speed:	0.00
-20	Y Speed:	0.00
-40	5s Avg:	2.12
-60	10s Avg:	1.18
-80		

Hold **L2** + **X** Super Slingshot



Lessons Learned

- The thing that feels right is the right thing to make

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- The thing that feels right is the right thing to make
- A feature is only good if a target player can and will use it
- Reward loops don't always have to be seen to be felt



Thank You

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SPIDER-MAN 2

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